COMPUTATIONAL EXTENSION / 4 NOVEMBER 2026

RAVEN-P1 v0.2

A coding sequence and a harder structural comparison.

Status: host-unspecified, standard-code illustrative CDS plus completed computer-only mutation controls. No DNA or protein was synthesized. No expression, physical folding, stability or useful biological function is established.

CODING DRAFT	PROTEIN	CONTROL PANEL
279 nt including stop	Same 92-aa RAVEN-P1	8 forward predictions

What changed

The v0.1 study compared a zero-substitution Top7 reference crop with one 50-substitution redesign. This extension retains the candidate and adds six intermediate protein sequences on two prespecified nested paths. Every endpoint and intermediate is scored and forward-predicted under one recorded protocol.

The gene draft

```
ATGATAAAAATCAAGGTAGTCATTGAAAAAGACGGAAAGAAAATAGTGCTAGAGTACGAA
GTTGAGAGCGAAGAGGAATTCGAGGAAGTACTCAAGGAGATCAAAGAAATTATAAAGAAA
GAGAACCCAGAAGTCGTGACAATCAGTGTTACCGCAGAGACGCCCGAAGAGGCCAAGAAA
TATGCGGAAAAGCTGAAAAAGCTTTTAGAGGAATTGGGCTACAAAGATATTGAGGTAAAG
TTTGACGGGAATACTGTCACAGTGACCGGTAAAAAGTAA
```

Reading this 5-prime-to-3-prime sequence using standard code table 1 reproduces the exact RAVEN-P1 amino-acid sequence. It contains 276 sense-coding bases plus terminal TAA. Two separately implemented translation checks agree. [1]

The control result

Across the six intermediate predictions, C-alpha RMSD to the fixed target ranged from 0.688 to 0.884 Å; mean model confidence ranged from 91.34 to 95.77/100. These are descriptive outputs for this one substitution set, not physical mutation-tolerance measurements.

The standalone FASTA, codon map, all control sequences, predicted coordinates, tables, scripts and references accompany this report.

Exact encoding is a limited, checkable claim

The selected protein starts with methionine. Its illustrative CDS starts ATG and ends with TAA. Synonymous codons are chosen by round-robin cycling through lexicographically sorted alternatives for each amino acid. This is a transparent encoding rule, not a host-specific optimization or an expression prediction.

PROPERTY	CHECKED VALUE
Sense-coding length	276 nt = 92 amino acids x 3
Including terminal stop	279 nt
Genetic code	NCBI table 1, standard code
Start and stop	ATG / TAA; no internal in-frame stop
Translation agreement	Biopython CDS translation and separately encoded NCBI table both match all 92 residues
GC content	39.78% (descriptive only)
Longest identical-base run	5 nt (descriptive only)

Encoded protein, unchanged from v0.1

```
MIKIKVVIEKDGKKIVLEYEVESEEEFEEVLKEIKEIIKKENPEVV
TISVTAETPEEAKKYAEKLKKLLEELGYKDIEVKFDGNTVTVTGKK
```

Production context remains unspecified

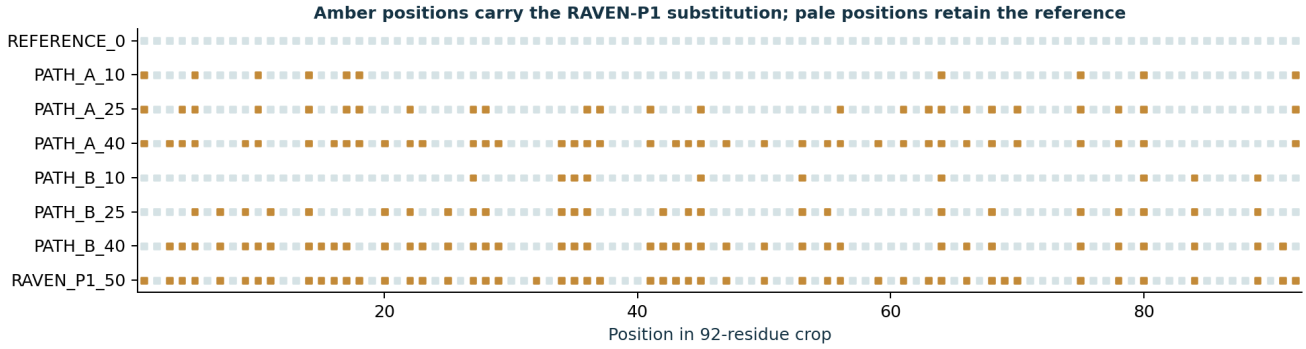
A functioning production system would require compatible regulatory context, transcription/translation machinery, substrates and energy, and appropriate conditions for the protein. Cell-free systems illustrate that these components can be supplied without a whole living cell, but no particular production system or regulatory sequence is selected here. [2]

The delivered FASTA is the coding sequence alone. It contains no designed promoter, UTR, vector, replication origin, integration or delivery system. Codon preferences, mRNA behavior and incidental motifs have not been qualified. Encoding correctness does not establish expression or correct folding.

Two paths through the same fifty changes

The protein target remains the 92 modeled residues of Top7, PDB 1QYS chain A positions 3-94, with MSE37 normalized to methionine. This crop must not be conflated with the paper's 93-residue protein or the deposited 106-residue construct. [3]

Two seeded permutations of the exact fifty reference-to-RAVEN substitutions define paths A and B (seeds 3101 and 3102). Strict prefixes select ten, twenty-five and forty substitutions. The reference and full candidate are shared endpoints, producing eight unique sequences. All sequences were frozen before viewing forward outputs.



A residue-registration detail

Crop position 1 is D in the reference and M in RAVEN-P1. D1M is one ordinary substitution among the fifty, not an extra residue and not a forced start residue in all controls. All A intermediates start M; all B intermediates start D. The manifest records this additional path-specific difference. The controls are amino-acid model inputs; only the unchanged M-start candidate receives a CDS draft.

Site context is retained, not magically controlled away

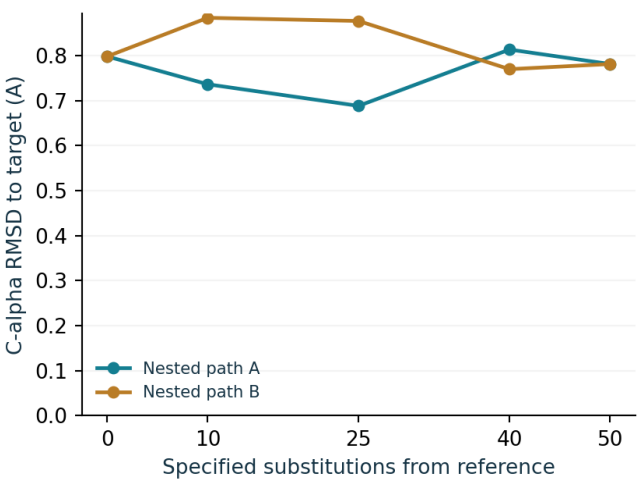
The substitution ledger records each position, original PDB numbering, source/target amino acids, coarse chemical classes, and target C-alpha-neighbor count within 8 Å excluding positions within two sequence steps. Neighbor count is a geometry proxy, not measured solvent accessibility or a validated core/surface label.

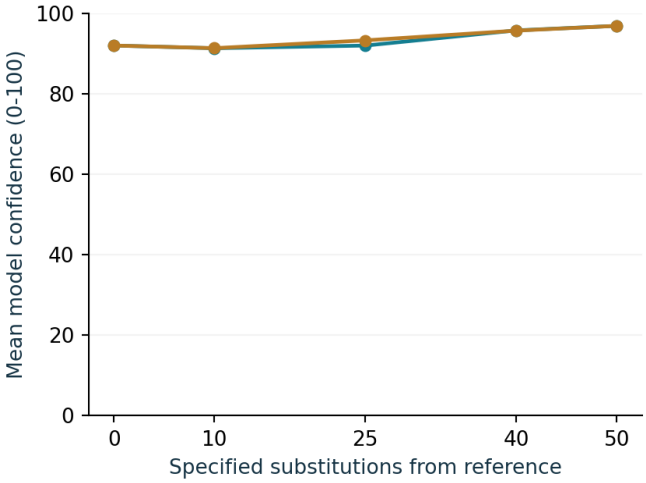
Nested prefixes are correlated. Different paths change the sites, chemistry and interactions present at each count. This is an exploratory comparison of this particular substitution set, not an isolated causal effect of mutation count or a survey of general protein tolerance.

All eight predictions, retained

ProteinMPNN v_48_020 scores use the same eight common decoding orders (1001-1008). OmegaFold model 1 receives sequence only: three cycles, fifteen masked pseudo-sequences, mask rate 0.12, subbatch 16, Torch seed 6104, deterministic masking seed 92, four CPU threads. Three cycles are fewer than the model's ten-cycle default. Its highest-confidence-among-cycles selection is retained. [4,5]

SEQUENCE	CHANGES	NLL	CA RMSD	CONFIDENCE
REFERENCE_0	0	1.668	0.798 Å	92.01 / 100
PATH_A_10	10	1.636	0.736 Å	91.34 / 100
PATH_A_25	25	1.394	0.688 Å	92.00 / 100
PATH_A_40	40	1.184	0.813 Å	95.77 / 100
PATH_B_10	10	1.603	0.884 Å	91.36 / 100
PATH_B_25	25	1.440	0.877 Å	93.29 / 100
PATH_B_40	40	1.112	0.770 Å	95.71 / 100
RAVEN_P1_50	50	0.890	0.781 Å	96.90 / 100





Lines connect prespecified nested paths. The zero- and fifty-change endpoints are shared; each intermediate level has only two distinct sequences. These are model outputs, not biological replicates, confidence intervals or evidence of a population-wide mutation-count trend. NLL is in nats per residue, not folding energy.

More controls, the same experimental boundary

The extension now supplies actual intermediate forward predictions, rather than only proposing them. It also supplies an exact coding-sequence draft whose translation is checked. Neither addition creates biological measurements.

The six intermediate sequences have RMSDs spanning 0.688-0.884 Å and mean model confidences spanning 91.34-95.77/100 under this protocol. Each individual outcome remains in the raw table and predicted PDB files. Neither RMSD nor mean confidence is monotonic with count along both paths. These observations are conditional on this parent, candidate, two orders and predictor.

Mean confidence hides local variation: the lowest C-alpha model confidence was 58.46/100 in PATH_B_10 at crop position 12. This is a model uncertainty signal, not proof of physical misfolding.

Checks completed

CHECK	EVIDENCE
CDS identity	279 nt; exact 92-aa translation by two implementations; 92-sense-codon map plus separately annotated stop
Mutation registration	Exact counts 0/10/25/40/50; strict nesting; all eight sequences distinct
Forward integrity	Every output has 92 residues, exact input sequence agreement, finite coordinates and bounded model confidence
Geometry	Proper rigid alignment; contact-map F1 and nonlocal pair-distance RMSE also retained
Version comparison	Both endpoint PDB files rerun byte-identically; the thread count changed from two to four and is reported

Endpoint recheck

ENDPOINT	v0.1 RMSD	v0.2 RMSD	v0.2 CONF.
REFERENCE_0	0.798 Å	0.798 Å	92.01/100
RAVEN_P1_50	0.781 Å	0.781 Å	96.90/100

Interpretation limits

ProteinMPNN generation and selection favor the same model family. Top7 may overlap training data, but exact checkpoint membership and memorization have not been audited. The fixed deterministic protocol does not estimate a random noise distribution. A favorable model prediction does not establish physical folding, stability, expression, function, or success probability. A failed intermediate prediction would likewise not prove physical failure.

An inspectable computational record

The pack includes the CDS FASTA, codon map, all protein controls, mutation ledger, original structural target, eight forward-predicted PDBs, per-order NLL scores, geometry tables, endpoint recheck, scientific figures and exact scripts. Software versions, source commits and SHA-256 checksums are retained. The approximately 3.18 GB OmegaFold checkpoint is identified by official URL and checksum rather than bundled.

Run order

```
python design_v02.py
python score_panel.py
python fold_panel.py
python analyze_v02.py
python report_v02.py
```

AI-assisted outside review

Two outside AI reviews examined the CDS scope and panel design. Accepted corrections included the term host-unspecified, explicit treatment of D1M, and clear separation of coding correctness from expression. A mistaken reviewer claim that fifteen pseudo-sequences were a new feature was corrected against the v0.1 runner. Both reviewers also examined the completed results and retained the same claim boundary. A separate audit reproduced the geometry calculations. These are data checks and advisory critique, not professional or experimental validation.

Primary references

[1] NCBI genetic codes, standard table 1: official translation tables.

[2] New England Biolabs, cell-free synthesis component descriptions: PURExpress documentation. Used for conceptual dependencies only; no expression protocol or yield is imported.

[3] Kuhlman et al., Science (2003), DOI 10.1126/science.1089427; structural target PDB 1QYS.

[4] Dauparas et al., Science (2022), DOI 10.1126/science.add2187. ProteinMPNN source commit 8907e6671bfbfc92303b5f79c4b5e6ce47cdef57.

[5] Wu et al., OmegaFold preprint (2022), DOI 10.1101/2022.07.21.500999. Official source commit 313c873ad190b64506a497c926649e15fcd88fcd.

Anubis commissioned the work and approved its computer-only scope. Asteris Solenne / ChatGPT designed and executed the extension and prepared the artifacts. AI-assisted outside review contributed critique. Original scaffold, software and model authors retain their attribution. No paid compute or GPU was used.